Atmospheric Histories (1765-2022) for CFC-11, CFC-12, CFC-113, CCl4, SF6, and N2O:
An update to Bullister (2015) NDP-09, *doi: 10.3334/CDIAC/otg.CFC_ATM_Hist_2015*

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Abstract

It has been several years since Bullister (2015) prepared tables and figures of the annual mean tropospheric concentrations in the northern (NH) and southern (SH) hemispheres for the trace gases: CFC-11, CFC-12, CFC-113, carbon tetrachloride (CCl4), sulfur hexafluoride (SF6), and nitrous oxide (N_2O). This is the first of periodic updates to the tables and figures compiled following similar methods with updated measurements from the community of scientists who monitor the global distributions of these compounds in the troposphere.

The datafile, Tracer_atmospheric_histories_revised_2023_Table1.csv, and Table 1 list the annual mean tropospheric concentrations of CFC-11, CFC-12, CFC-113, CCl_4, SF_6, and N_2O in the northern and southern hemispheres (NH, SH) from 1765.5 to 2022.5. The reported values are the annual (Jan-Dec) mean mole fraction of these gases in dry air. Concentrations are in parts per trillion (ppt) for CFC-11, CFC-12, CFC-113, CCl_4, and SF_6; and in parts per billion (ppb) for N_2O.

1. Details for CFC-11, CFC-12, CFC-113 and CCl_4

The concentrations for CFC-11, CFC-12, CFC-113, and CCl_4 listed in Table 1 for 1910.5 – 2008.5 are from the 2009 update of the Walker et al. (2000) results found at bluemoon.ucsd.edu/pub/cfchist. The concentrations for these trace gases from 2009.5 to 2022.5 are derived from the NOAA gas sampling program formerly known as CMDL, now known as GML (Global Monitoring Laboratory). These NOAA data are tabulated through Feb. 2023 (as of June 2023) and can be located in the subdirectories of https://gml.noaa.gov/aftp/data/hats/. Following the procedure of Bullister (2017), the changes in the annual average (Jan-Dec) concentrations of the trace gases in ppt were added to the Walker et al. data.
The sum of the changes in concentrations between 2008.5 and 2019.5 in the GML data set are slightly different than the differences in the global mean concentrations reported by the AGAGE measurement program (https://agage2.eas.gatech.edu/data_archive/global_mean/) which may indicate a small systematic offset. These differences are consistent with those reported by Rigby et al. (2013) for the earlier part of this time period.

Gas	Delta AGAGE	Delta GML (NH)	Delta GML (SH)
CFC-11	-17.33	-16.30	-16.62
CFC-12	-32.65	-34.80	-33.51
CCl_4	-11.43	-12.34	-12.08
CFC-113	-6.68	-6.90	-6.84

Details for SF_6

Bullister (2017) normalized the cumulative production and release estimates of this compound to the measured atmospheric concentrations of this tracer in the Northern and Southern Hemispheres in June 1995

by the NOAA gas sampling program – currently found at https://gml.noaa.gov/aftp/hats/sf6/combined/HATS_global_SF6.txt.

In Table 1, the SF_6 values from 1996.5 to 2022.5 are the mean annual NH and SH values reported by GML. Note that the GML concentrations are now reported on the NOAA 2014 scale, instead of the CMDL2006 scale as in Bullister (2017). Although the conversion between the two NOAA scales requires solving a quadratic equation (see https://gml.noaa.gov/ccl/sf6_scale.html) to revert to the CMDL 2006 scale, Krummell et al. (2017) reported that NOAA 2014/ SIO 2005 = 1.002 (± 0.002). The SF_6 data in Table 1 are reported on the SIO2005 scale using this conversion factor.

The AGAGE program reports SF_6 concentrations beginning in 2004. A comparison of the increase in reported atmospheric concentrations between 2004.5 and 2020.5 differ by less than 0.20 ppt between AGAGE (increase of 4.88 ppt in 16 y) and GML (increase of 4.76 ppt (NH) and 4.68 ppt (SH) in 16 y).

Details for N_2O

Bullister (2017) fitted a linear non-polynomial function to the combined ice core measurements (1502-1996) and CMDL atmospheric measurements (1978-2015) of N_2O concentrations. In the plots of the residuals of the data to this fit (Figs. 10 and 11, Bullister 2017), there appears to be a slight positive bias in the most recent atmospheric measurements.

In this update, the N2O concentrations reported in Table 1 after 2010.5 are the annual averages from the GML data (https://gml.noaa.gov/aftp/data/hats/n2o/combined/), although the data prior to 2010.5 are calculated from the fitted data through 2015.5. The difference in N_2O values in Table 1 from those in Table 1 of Bullister (2017) for 2010.5 to 2015.5 are (in ppb):

Year	W-B No. Hem	W-B So. Hem
2010.5	-0.28	-0.46
2011.5	-0.12	-0.09
2012.5	-0.32	-0.15
2013.5	-0.33	-0.20
2014.5	-0.03	-0.11
2015.5	-0.03	0.07

The N_2O data are reported on the NOAA 2006A scale

Additional References:

Combined Chlorofluorcarbon-11 data from the NOAA/ESRL Global Monitoring Division.

Combined Chlorofluorcarbon-12 data from the NOAA/ESRL Global Monitoring Division.

Combined Chlorofluorcarbon-113 data from the NOAA/ESRL Global Monitoring Division.

Combined sulfur hexafluoride data from the NOAA/ESRL Global Monitoring Division.

Combined nitrous oxide data from the NOAA/ESRL Global Monitoring Division.

Krummel, P., Montzka, S., Harth, C., Miller, B., Mühle, J., Dlugokencky, E., Salameh, P., Hall, B., O'Doherty, S., Steele, L., Dutton, G., Young, D., Nance, J., Langenfelds, R., Elkins, J., Loh, Z., Lunder, C., Fraser, P., Derek, N., Mitrevski, B., Weiss, R., and Prinn, R.: An update of comparisons of non-CO_2 trace gas measurements between AGAGE and NOAA at common sites, in: 19th WMO/IAEA Meeting on Carbon Dioxide, Other Greenhouse Gases, and Related Measurement Techniques (GGMT-2017), Dübendorf, Switzerland, available at: http://www.wmo.int/pages/prog/arep/gaw/documents/GGMT2017_T03_Krummel.pdf (last access: 16 January 2018), 2017.

Rigby, M., Prinn, R. G., O'Doherty, S., Montzka, S. A., McCulloch, A., Harth, C. M., Mühle, J., Salameh, P. K., Weiss, R. F., Young, D., Simmonds, P.G., Hall, B. D., Dutton, G.S., Nance, D., Mondeel, D.J., Elkins, J. W., Krummel, P. B., Steele, L.P., and Fraser, P.J. (2013) Re-evaluation of the lifetimes of the major CFCs and CH_3CCl_3 using atmospheric trends. Atmos. Chem. Phys., 13, 2691-2702. doi: *10.5194/acp-13-2691-2013*

APPENDIX 1

Table 1. Annual mean tropospheric concentrations of CFC-11, CFC-12, CFC-113, CCl_4, SF_6, and N_2O in the northern and southern hemispheres (NH, SH) from 1765.5 to 2022.5. The reported values are the annual (Jan-Dec) mean mole fraction of these gases in dry air. Concentrations are in parts per trillion (ppt) for CFC-11, CFC-12, CFC-113, CCl_4, and SF_6; and in parts per billion (ppb) for N_2O. CFC-11, CFC-12, CFC-113, and CCl_4 concentrations are reported on the SIO 1998 scale; SF_6 is reported on the SIO2005 scale; N_2O concentrations are reported on the NOAA2006A scale.

YEAR	CFC11 NH ppt	CFC11 SH ppt	CFC12 NH ppt	CFC12 SH ppt	CFC113 NH ppt	CFC113 SH ppt	CCl4 NH ppt	CCl4 SH ppt	SF6 NH ppt	SF6 SH ppt	N2O NH ppb	N2O SH ppb
1765.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.47	270.47
1766.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.49	270.49
1767.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.50	270.50
1768.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.52	270.52
1769.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.53	270.53
1770.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.55	270.55
1771.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.56	270.56
1772.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.58	270.58
1773.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.59	270.59

1774.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.61	270.61
1775.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.62	270.62
1776.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.64	270.64
1777.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.66	270.66
1778.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.67	270.67
1779.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.69	270.69
1780.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.71	270.71
1781.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.73	270.73
1782.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.74	270.74
1783.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.76	270.76
1784.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.78	270.78
1785.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.80	270.80
1786.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.82	270.82
1787.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.84	270.84
1788.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.86	270.86
1789.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.88	270.88
1790.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.90	270.90
1791.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.92	270.92
1792.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.95	270.95
1793.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.97	270.97
1794.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	270.99	270.99
1795.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.02	271.02
1796.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.04	271.04
1797.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.06	271.06
1798.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.09	271.09
1799.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.11	271.11
1800.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.14	271.14
1801.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.16	271.16
1802.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.19	271.19
1803.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.21	271.21
1804.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.24	271.24
1805.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.27	271.27
1806.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.30	271.30
1807.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.33	271.33
1808.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.36	271.36
1809.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.39	271.39
1810.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.42	271.42
1811.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.45	271.45
1812.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.48	271.48
1813.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.51	271.51
1814.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.54	271.54
1815.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.57	271.57
1816.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.61	271.61
1817.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.64	271.64
1818.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.67	271.67

1819.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.71	271.71
1820.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.75	271.75
1821.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.78	271.78
1822.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.82	271.82
1823.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.86	271.86
1824.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.89	271.89
1825.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.93	271.93
1826.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	271.97	271.97
1827.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.01	272.01
1828.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.05	272.05
1829.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.10	272.10
1830.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.14	272.14
1831.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.18	272.18
1832.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.23	272.23
1833.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.27	272.27
1834.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.32	272.32
1835.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.36	272.36
1836.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.41	272.41
1837.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.46	272.46
1838.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.51	272.51
1839.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.56	272.56
1840.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.61	272.61
1841.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.66	272.66
1842.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.71	272.71
1843.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.77	272.77
1844.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.82	272.82
1845.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.88	272.88
1846.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.93	272.93
1847.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	272.99	272.99
1848.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	273.05	273.05
1849.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	273.11	273.11
1850.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	273.17	273.17
1851.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	273.23	273.23
1852.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	273.29	273.29
1853.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	273.35	273.35
1854.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	273.42	273.42
1855.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	273.48	273.48
1856.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	273.55	273.55
1857.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	273.62	273.61
1858.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	273.69	273.68
1859.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	273.76	273.75
1860.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	273.83	273.82
1861.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	273.91	273.89
1862.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	273.98	273.97
1863.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	274.06	274.04

1864.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	274.14	274.12
1865.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	274.22	274.19
1866.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	274.30	274.27
1867.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	274.38	274.35
1868.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	274.46	274.43
1869.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	274.55	274.51
1870.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	274.63	274.60
1871.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	274.72	274.68
1872.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	274.81	274.77
1873.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	274.90	274.86
1874.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	274.99	274.95
1875.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	275.08	275.04
1876.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	275.18	275.13
1877.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	275.27	275.23
1878.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	275.37	275.32
1879.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	275.47	275.42
1880.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	275.58	275.52
1881.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	275.68	275.62
1882.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	275.79	275.72
1883.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	275.89	275.83
1884.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	276.00	275.93
1885.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	276.11	276.04
1886.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	276.23	276.15
1887.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	276.34	276.26
1888.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	276.46	276.38
1889.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	276.58	276.49
1890.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	276.70	276.61
1891.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	276.82	276.73
1892.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	276.95	276.85
1893.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	277.07	276.98
1894.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	277.20	277.10
1895.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	277.33	277.23
1896.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	277.47	277.36
1897.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	277.60	277.49
1898.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	277.74	277.63
1899.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	277.88	277.77
1900.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	278.03	277.91
1901.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	278.17	278.05
1902.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	278.32	278.20
1903.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	278.47	278.34
1904.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	278.62	278.49
1905.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	278.78	278.64
1906.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	278.94	278.80
1907.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	279.10	278.96
1908.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	279.26	279.12

1909.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	279.43	279.28
1910.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	279.60	279.44
1911.5	0.00	0.00	0.00	0.00	0.00	0.00	0.10	0.00	0.00	0.00	279.77	279.61
1912.5	0.00	0.00	0.00	0.00	0.00	0.00	0.10	0.10	0.00	0.00	279.95	279.78
1913.5	0.00	0.00	0.00	0.00	0.00	0.00	0.10	0.10	0.00	0.00	280.13	279.96
1914.5	0.00	0.00	0.00	0.00	0.00	0.00	0.20	0.10	0.00	0.00	280.31	280.14
1915.5	0.00	0.00	0.00	0.00	0.00	0.00	0.30	0.20	0.00	0.00	280.49	280.32
1916.5	0.00	0.00	0.00	0.00	0.00	0.00	0.50	0.30	0.00	0.00	280.68	280.50
1917.5	0.00	0.00	0.00	0.00	0.00	0.00	0.80	0.50	0.00	0.00	280.87	280.69
1918.5	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.70	0.00	0.00	281.07	280.88
1919.5	0.00	0.00	0.00	0.00	0.00	0.00	1.10	0.90	0.00	0.00	281.26	281.07
1920.5	0.00	0.00	0.00	0.00	0.00	0.00	1.20	1.10	0.00	0.00	281.46	281.27
1921.5	0.00	0.00	0.00	0.00	0.00	0.00	1.30	1.20	0.00	0.00	281.67	281.46
1922.5	0.00	0.00	0.00	0.00	0.00	0.00	1.40	1.20	0.00	0.00	281.88	281.67
1923.5	0.00	0.00	0.00	0.00	0.00	0.00	1.70	1.40	0.00	0.00	282.09	281.87
1924.5	0.00	0.00	0.00	0.00	0.00	0.00	1.90	1.60	0.00	0.00	282.30	282.08
1925.5	0.00	0.00	0.00	0.00	0.00	0.00	2.20	1.80	0.00	0.00	282.52	282.30
1926.5	0.00	0.00	0.00	0.00	0.00	0.00	2.50	2.10	0.00	0.00	282.74	282.52
1927.5	0.00	0.00	0.00	0.00	0.00	0.00	2.80	2.40	0.00	0.00	282.97	282.74
1928.5	0.00	0.00	0.00	0.00	0.00	0.00	3.10	2.70	0.00	0.00	283.20	282.96
1929.5	0.00	0.00	0.00	0.00	0.00	0.00	3.60	3.00	0.00	0.00	283.43	283.19
1930.5	0.00	0.00	0.00	0.00	0.00	0.00	4.20	3.50	0.00	0.00	283.67	283.42
1931.5	0.00	0.00	0.00	0.00	0.00	0.00	4.80	4.10	0.00	0.00	283.91	283.66
1932.5	0.00	0.00	0.00	0.00	0.00	0.00	5.40	4.60	0.00	0.00	284.16	283.90
1933.5	0.00	0.00	0.00	0.00	0.00	0.00	6.00	5.20	0.00	0.00	284.41	284.14
1934.5	0.00	0.00	0.00	0.00	0.00	0.00	6.80	5.80	0.00	0.00	284.66	284.39
1935.5	0.00	0.00	0.00	0.00	0.00	0.00	7.80	6.60	0.00	0.00	284.92	284.64
1936.5	0.00	0.00	0.10	0.00	0.00	0.00	8.90	7.50	0.00	0.00	285.18	284.90
1937.5	0.00	0.00	0.10	0.10	0.00	0.00	10.30	8.60	0.00	0.00	285.45	285.16
1938.5	0.00	0.00	0.20	0.10	0.00	0.00	11.60	9.90	0.00	0.00	285.72	285.43
1939.5	0.00	0.00	0.30	0.20	0.00	0.00	12.90	11.10	0.00	0.00	286.00	285.70
1940.5	0.00	0.00	0.40	0.30	0.00	0.00	14.40	12.40	0.00	0.00	286.28	285.97
1941.5	0.00	0.00	0.50	0.40	0.00	0.00	16.20	14.00	0.00	0.00	286.56	286.25
1942.5	0.00	0.00	0.70	0.50	0.00	0.00	18.50	15.80	0.00	0.00	286.86	286.54
1943.5	0.00	0.00	0.90	0.70	0.00	0.00	21.20	18.00	0.00	0.00	287.15	286.83
1944.5	0.00	0.00	1.20	0.90	0.00	0.00	24.20	20.50	0.00	0.00	287.45	287.12
1945.5	0.10	0.00	1.70	1.20	0.00	0.00	26.70	23.20	0.00	0.00	287.76	287.42
1946.5	0.10	0.10	2.30	1.70	0.00	0.00	28.20	25.30	0.00	0.00	288.07	287.73
1947.5	0.10	0.10	3.40	2.40	0.00	0.00	30.10	27.10	0.00	0.00	288.38	288.04
1948.5	0.20	0.10	4.80	3.40	0.00	0.00	32.50	29.20	0.00	0.00	288.71	288.35
1949.5	0.40	0.20	6.10	4.70	0.00	0.00	34.20	31.20	0.00	0.00	289.03	288.67
1950.5	0.70	0.40	7.60	6.00	0.00	0.00	35.60	32.90	0.00	0.00	289.36	288.99
1951.5	1.01	0.70	9.20	7.40	0.00	0.00	37.30	34.40	0.00	0.00	289.70	289.33
1952.5	1.51	1.01	11.00	9.00	0.00	0.00	38.40	35.90	0.00	0.00	290.05	289.66
1953.5	2.21	1.51	12.80	10.70	0.00	0.00	39.30	37.00	0.04	0.04	290.40	290.01

1954.5	3.02	2.21	15.00	12.60	0.00	0.00	39.70	37.80	0.04	0.04	290.75	290.35
1955.5	4.12	3.02	17.40	14.70	0.00	0.00	40.10	38.30	0.04	0.04	291.11	290.71
1956.5	5.33	4.02	20.20	17.10	0.00	0.00	40.40	38.70	0.04	0.04	291.48	291.07
1957.5	6.83	5.23	23.40	19.90	0.00	0.00	40.60	39.00	0.05	0.04	291.86	291.43
1958.5	8.14	6.53	26.80	23.00	0.00	0.00	41.50	39.50	0.05	0.04	292.23	291.80
1959.5	9.45	7.84	30.50	26.30	0.00	0.00	42.60	40.30	0.05	0.05	292.62	292.18
1960.5	11.06	9.15	35.00	30.10	0.00	0.00	43.70	41.20	0.05	0.05	293.01	292.57
1961.5	13.27	10.85	40.00	34.40	0.10	0.10	44.70	42.20	0.06	0.06	293.41	292.96
1962.5	16.18	13.07	45.80	39.40	0.30	0.10	45.90	43.30	0.07	0.06	293.82	293.36
1963.5	19.60	15.78	52.50	45.10	0.50	0.30	47.60	44.50	0.08	0.07	294.23	293.76
1964.5	23.72	19.20	60.40	51.80	0.80	0.50	49.40	46.10	0.09	0.08	294.65	294.17
1965.5	28.44	23.12	69.30	59.50	1.10	0.80	50.90	47.70	0.11	0.10	295.08	294.59
1966.5	33.67	27.64	79.20	68.20	1.49	1.10	53.00	49.30	0.13	0.12	295.52	295.02
1967.5	39.40	32.66	90.30	77.90	1.99	1.49	55.80	51.40	0.15	0.14	295.96	295.45
1968.5	46.03	38.29	102.80	88.80	2.69	1.99	59.10	54.10	0.18	0.17	296.41	295.89
1969.5	53.77	44.82	116.80	101.10	3.39	2.59	63.40	57.40	0.21	0.19	296.86	296.34
1970.5	62.41	52.26	132.00	114.70	4.18	3.29	69.30	61.70	0.23	0.22	297.33	296.79
1971.5	72.06	60.70	148.40	129.60	5.18	4.08	74.90	67.00	0.26	0.24	297.80	297.26
1972.5	82.71	69.95	166.10	145.70	6.27	5.08	77.60	71.40	0.30	0.28	298.28	297.73
1973.5	94.87	80.40	185.80	163.30	7.47	6.18	79.70	74.30	0.34	0.31	298.77	298.21
1974.5	108.34	92.16	207.10	182.50	8.96	7.37	83.20	77.20	0.38	0.35	299.26	298.69
1975.5	121.41	104.72	228.20	202.90	10.66	8.76	85.20	80.00	0.44	0.40	299.77	299.19
1976.5	133.97	117.09	248.10	223.20	12.45	10.36	87.00	82.10	0.50	0.46	300.28	299.69
1977.5	145.93	129.35	266.90	242.70	14.64	12.25	89.80	84.40	0.58	0.53	300.80	300.20
1978.5	156.58	140.80	284.30	261.20	16.93	14.34	91.90	86.70	0.66	0.61	301.33	300.72
1979.5	168.34	148.74	306.10	273.50	19.62	16.63	93.30	88.60	0.76	0.70	301.87	301.25
1980.5	176.68	159.30	323.20	292.30	22.61	19.32	94.90	89.60	0.88	0.81	302.42	301.79
1981.5	184.32	167.84	339.60	308.80	25.90	22.21	96.40	90.80	1.00	0.93	302.98	302.33
1982.5	191.46	176.08	353.40	325.50	29.18	25.30	97.50	92.30	1.13	1.04	303.55	302.89
1983.5	199.30	184.52	369.00	342.60	32.97	28.09	98.50	93.50	1.27	1.17	304.12	303.46
1984.5	208.04	192.46	385.70	359.40	37.85	31.37	99.80	95.10	1.40	1.29	304.71	304.03
1985.5	217.99	202.01	403.40	378.20	43.92	35.96	100.50	96.60	1.55	1.43	305.30	304.62
1986.5	229.35	211.36	424.30	396.50	48.31	40.94	101.90	98.20	1.71	1.58	305.91	305.21
1987.5	241.61	222.21	444.00	416.30	55.68	47.01	102.50	100.00	1.88	1.73	306.53	305.82
1988.5	252.86	233.27	465.40	435.80	63.55	53.29	103.50	100.60	2.05	1.89	307.15	306.43
1989.5	259.30	242.11	483.60	454.40	69.72	59.66	104.40	101.10	2.22	2.05	307.79	307.05
1990.5	265.83	251.06	497.70	472.70	75.60	66.14	104.90	102.30	2.41	2.22	308.44	307.69
1991.5	268.24	256.68	506.00	487.30	80.98	71.31	105.50	102.50	2.62	2.42	309.10	308.34
1992.5	268.14	260.80	516.30	498.30	83.96	77.19	104.90	101.90	2.85	2.63	309.77	308.99
1993.5	269.55	262.51	523.20	507.00	84.36	79.58	104.00	101.10	3.09	2.84	310.45	309.66
1994.5	269.65	263.72	528.50	514.80	84.46	81.27	103.00	100.30	3.33	3.07	311.14	310.34
1995.5	268.34	263.22	533.40	521.00	84.36	82.17	102.10	99.60	3.55	3.30	311.84	311.03
1996.5	266.93	262.91	537.30	526.50	84.06	82.67	101.10	98.60	3.82	3.55	312.56	311.73
1997.5	265.73	262.01	540.10	530.80	83.57	82.67	100.20	98.00	4.05	3.81	313.29	312.45
1998.5	264.52	261.01	542.90	534.30	83.17	82.47	99.20	97.00	4.26	4.01	314.03	313.17

1999.5	263.12	259.90	544.40	537.20	82.47	81.97	98.20	96.20	4.46	4.25	314.78	313.91
2000.5	261.71	258.29	545.90	539.00	81.87	81.37	97.30	95.40	4.65	4.46	315.54	314.66
2001.5	260.00	256.98	546.50	540.60	81.18	80.78	96.30	94.40	4.86	4.65	316.32	315.42
2002.5	258.19	255.08	546.70	541.30	80.48	79.98	95.30	93.60	5.11	4.86	317.11	316.20
2003.5	256.18	253.27	546.70	541.60	79.88	79.38	94.30	92.50	5.35	5.08	317.91	316.99
2004.5	253.97	251.36	545.70	541.50	79.08	78.78	93.30	91.70	5.58	5.31	318.73	317.79
2005.5	251.96	249.15	544.90	540.70	78.49	78.09	92.40	90.80	5.80	5.53	319.56	318.60
2006.5	249.55	247.34	543.10	539.80	77.69	77.59	91.30	89.70	6.04	5.76	320.41	319.43
2007.5	247.54	245.03	541.10	538.10	76.99	76.79	90.40	88.90	6.32	6.03	321.26	320.28
2008.5	245.63	243.12	538.60	536.20	76.29	76.20	89.30	87.70	6.62	6.30	322.14	321.13
2009.5	243.78	241.29	536.02	533.71	75.75	75.61	87.90	86.41	6.91	6.59	323.02	322.00
2010.5	241.94	239.46	532.46	530.20	75.10	75.02	86.97	85.44	7.21	6.87	323.65	322.43
2011.5	239.89	237.36	529.72	527.58	74.43	74.26	85.62	84.11	7.47	7.15	324.72	323.70
2012.5	237.83	235.40	527.00	524.98	73.73	73.57	84.44	83.01	7.76	7.43	325.46	324.56
2013.5	236.36	233.52	523.48	521.72	73.02	72.92	83.42	81.95	8.09	7.73	326.40	325.44
2014.5	235.38	231.81	520.77	518.62	72.41	72.29	82.46	80.63	8.43	8.06	327.66	326.46
2015.5	234.14	230.65	517.93	515.57	71.84	71.70	81.24	79.33	8.76	8.40	328.64	327.62
2016.5	233.18	229.64	514.76	512.94	71.26	71.15	80.08	78.24	9.10	8.72	329.55	328.32
2017.5	232.30	228.58	511.08	509.63	70.69	70.53	79.19	77.11	9.44	9.06	330.32	329.18
2018.5	231.27	228.08	508.06	507.01	70.09	70.02	77.97	76.23	9.78	9.42	331.35	330.38
2019.5	229.33	226.51	503.82	502.69	69.40	69.36	76.97	75.62	10.12	9.76	332.34	331.37
2020.5	226.58	224.24	499.81	498.39	68.77	68.68	75.73	74.50	10.44	10.08	333.71	332.40
2021.5	224.01	222.19	495.61	494.81	68.14	68.10	74.61	73.50	10.84	10.44	334.85	333.82
2022.5	221.71	220.08	491.93	491.46	67.56	67.58	73.23	72.58	11.23	10.82	336.15	335.27

Appendix 2 (Figures)

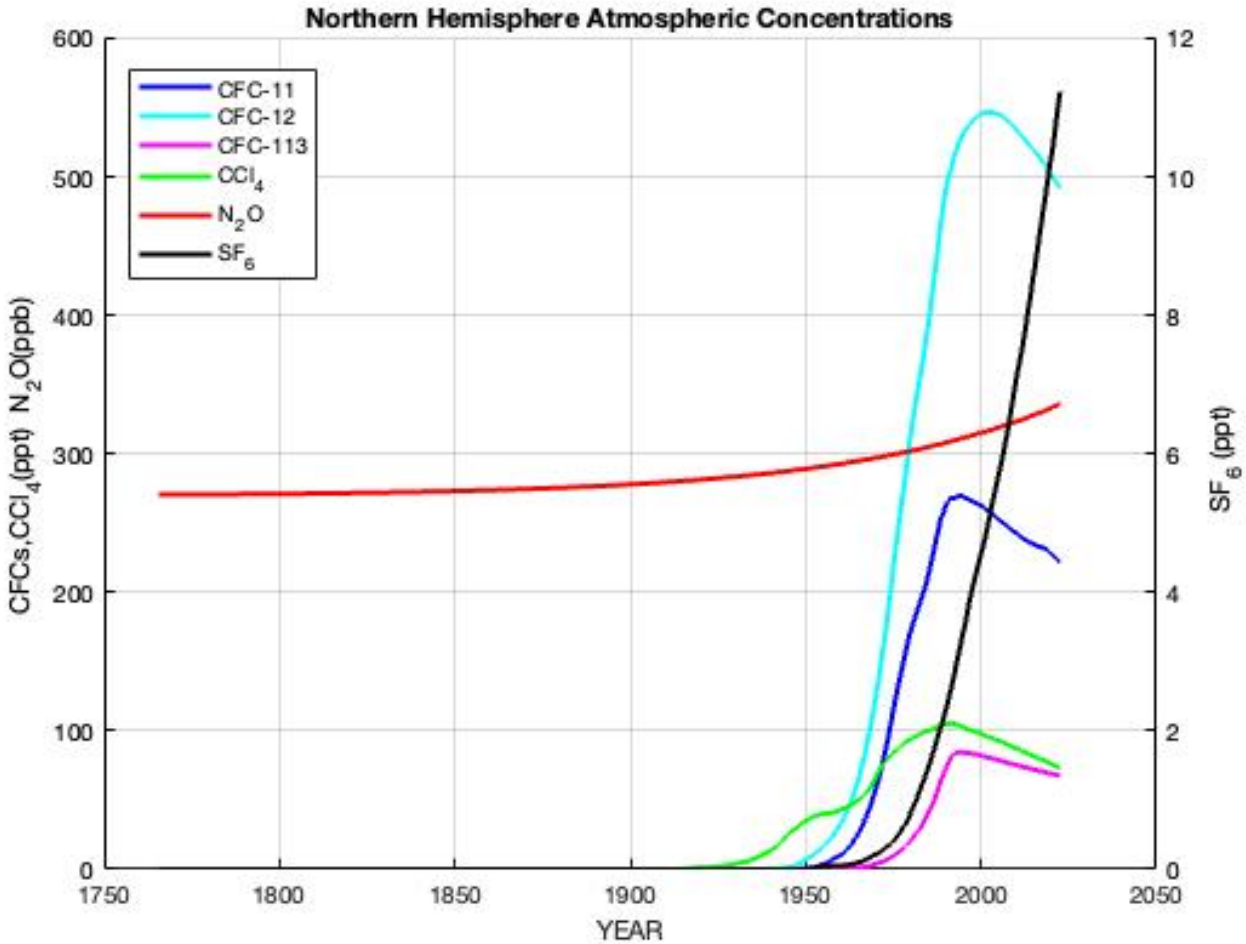

Fig.1. Mean annual tropospheric CFC-11, CFC-12, CFC-113, carbon tetrachloride (CCl_4), sulfur hexafluoride (SF_6) and nitrous oxide (N_2O) concentrations in the northern (NH) hemisphere for the period 1765.5 to 2022.5. The concentrations are expressed as the mixing ratio (mole fraction) of the trace gas in dry air and are reported in parts-per-trillion (ppt) for CFC-11, CFC-12, CFC-113, CCl_4, SF_6 and as parts-per-billion (ppb) for N_2O.

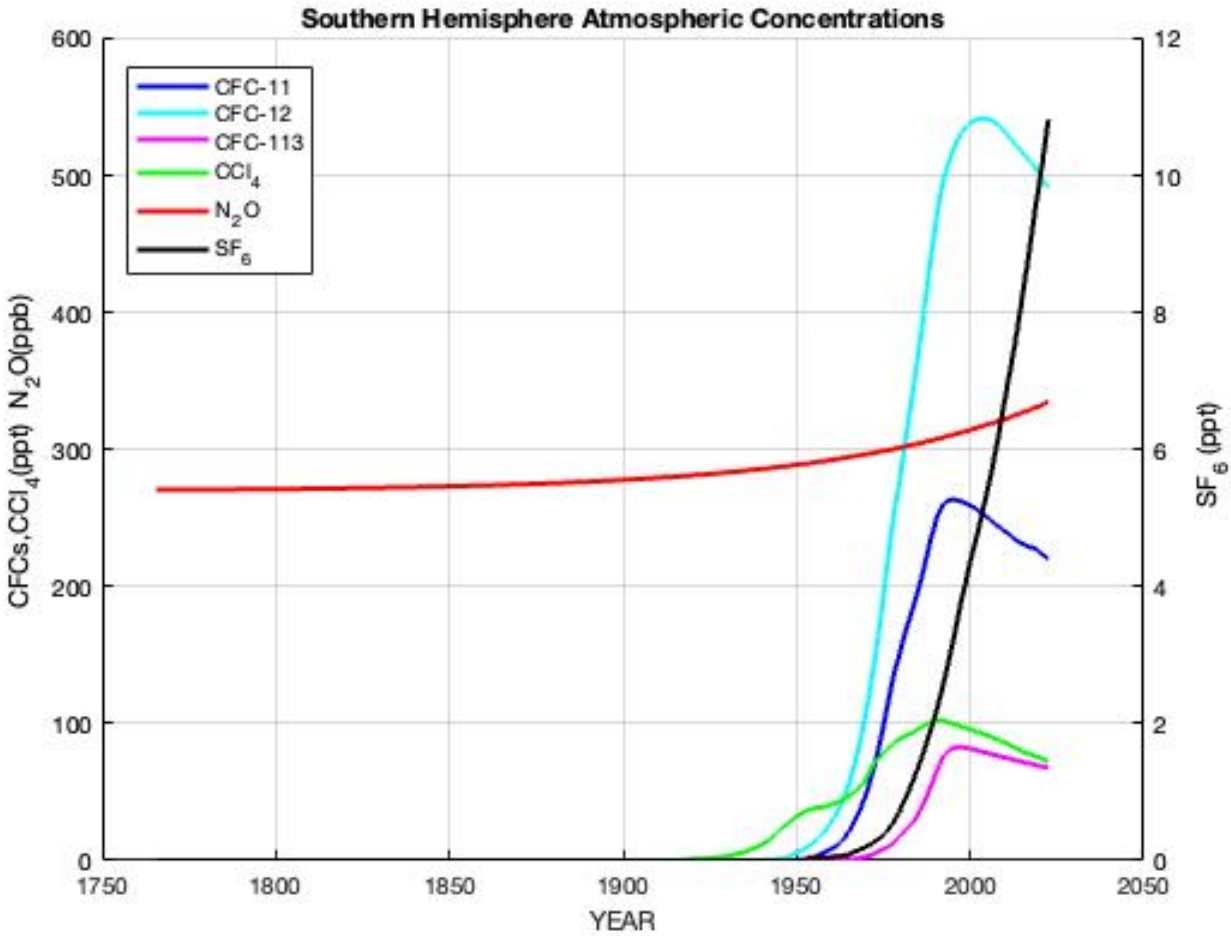

Fig. 2. Mean annual tropospheric CFC-11, CFC-12, CFC-113, carbon tetrachloride (CCl_4), sulfur hexafluoride (SF_6) and nitrous oxide (N_2O) concentrations in the southern (SH) hemisphere for the period 1765.5 to 2022.5. The concentrations are expressed as the mixing ratio (mole fraction) of the trace gas in dry air and are reported in parts-per-trillion (ppt) for CFC-11, CFC-12, CFC-113, CCl_4, SF_6 and as parts-per-billion (ppb) for N_2O.

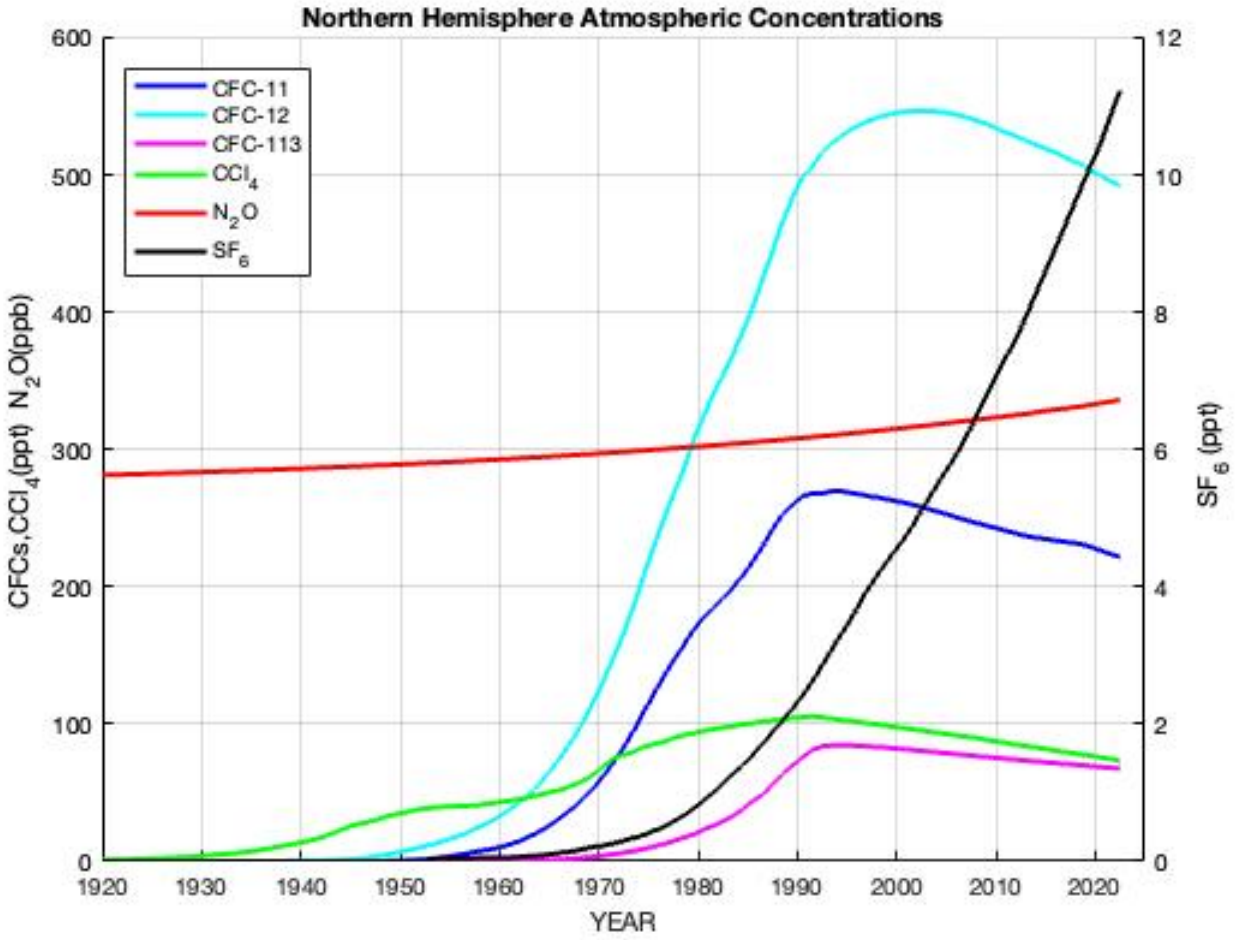

Fig. 3. Mean annual tropospheric CFC-11, CFC-12, CFC-113, carbon tetrachloride (CCl_4), sulfur hexafluoride (SF_6) and nitrous oxide (N_2O) concentrations in the northern (NH) hemisphere for the period 1920.5 to 2022.5. The concentrations are expressed as the mixing ratio (mole fraction) of the trace gas in dry air and are reported in parts-per-trillion (ppt) for CFC-11, CFC-12, CFC-113, CCl_4, SF_6 and as parts-per-billion (ppb) for N_2O.

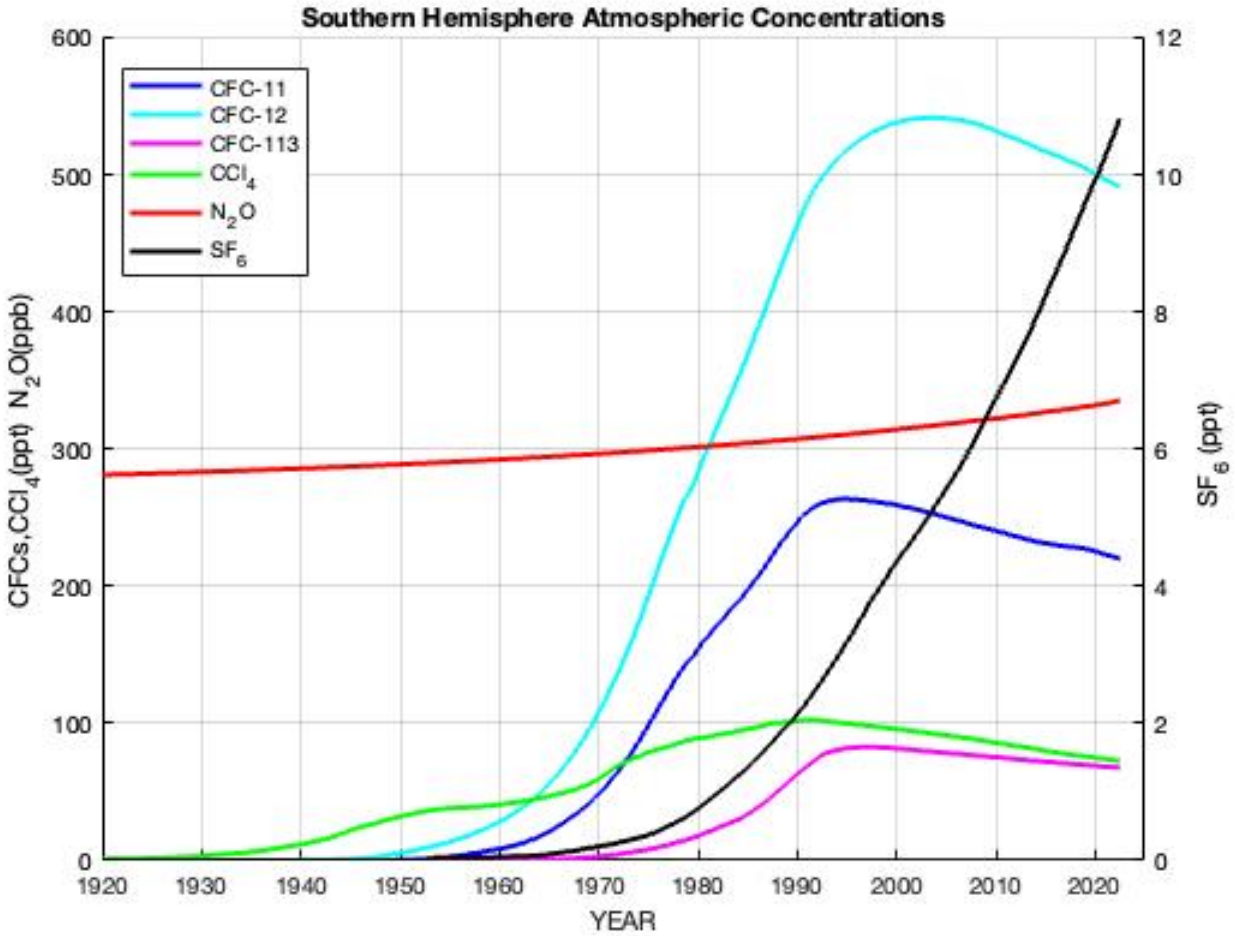

Fig. 4. Mean annual tropospheric CFC-11, CFC-12, CFC-113, carbon tetrachloride (CCl_4), sulfur hexafluoride (SF_6) and nitrous oxide (N_2O) concentrations in the southern (SH) hemisphere for the period 1920.5 to 2022.5. The concentrations are expressed as the mixing ratio (mole fraction) of the trace gas in dry air and are reported in parts-per-trillion (ppt) for CFC-11, CFC-12, CFC-113, CCl_4, SF_6 and as parts-per-billion (ppb) for N_2O.

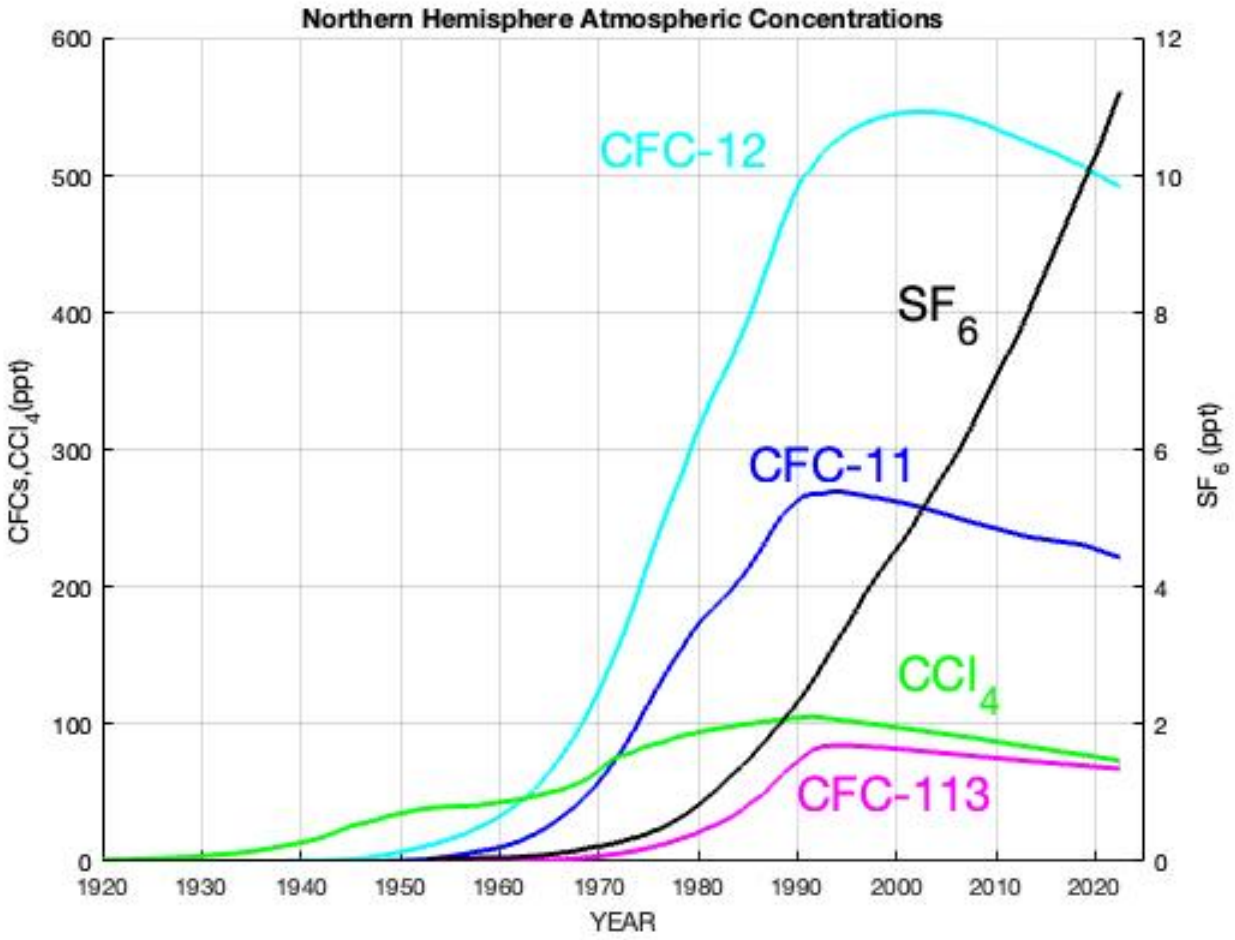

Fig. 5. Mean annual tropospheric CFC-11, CFC-12, CFC-113, carbon tetrachloride (CCl_4) and sulfur hexafluoride (SF_6) concentrations in the northern (NH) hemisphere for the period 1920.5 to 2022.5. The concentrations are expressed as the mixing ratio (mole fraction) of the trace gas in dry air and are reported in parts-per-trillion (ppt).

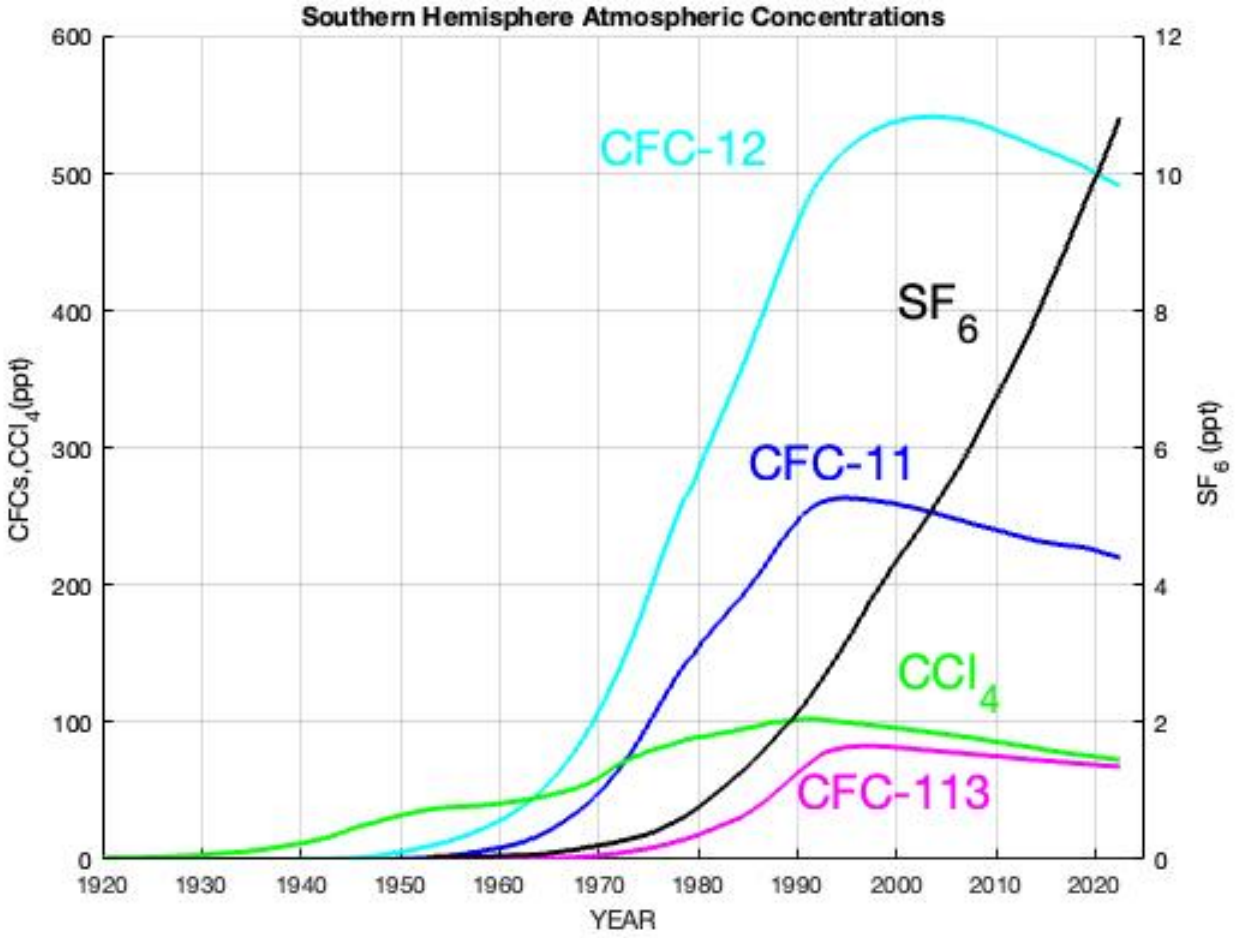

Fig. 6. Mean annual tropospheric CFC-11, CFC-12, CFC-113, carbon tetrachloride (CCl_4) and sulfur hexafluoride (SF_6) concentrations in the southern (SH) hemisphere for the period 1920.5 to 2022.5. The concentrations are expressed as the mixing ratio (mole fraction) of the trace gas in dry air and are reported in parts-per-trillion (ppt).

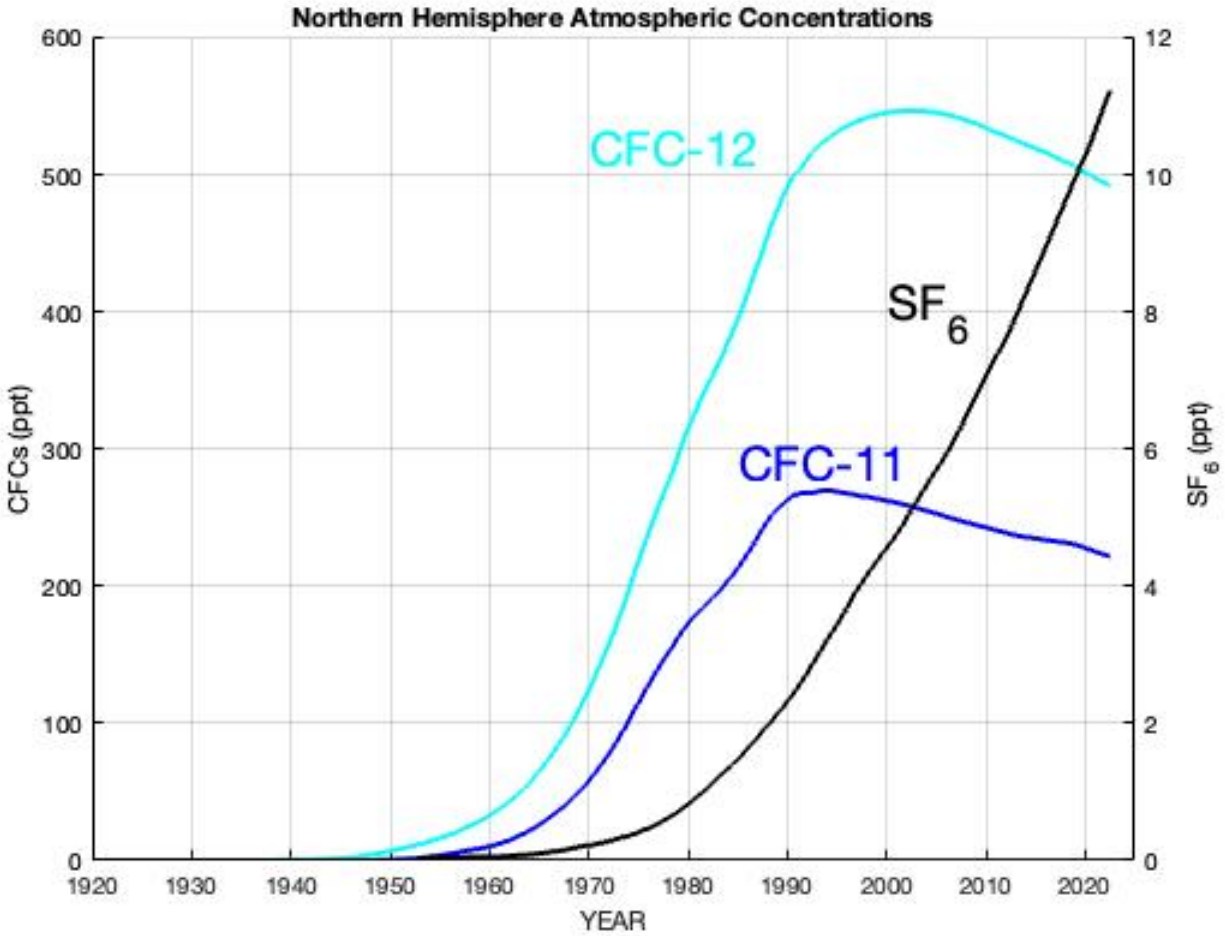

Fig. 7. Mean annual tropospheric CFC-11, CFC-12 and sulfur hexafluoride (SF_6) concentrations in the northern (NH) hemisphere for the period 1920.5 to 2022.5. The concentrations are expressed as the mixing ratio (mole fraction) of the trace gas in dry air and are reported in parts-per-trillion (ppt).

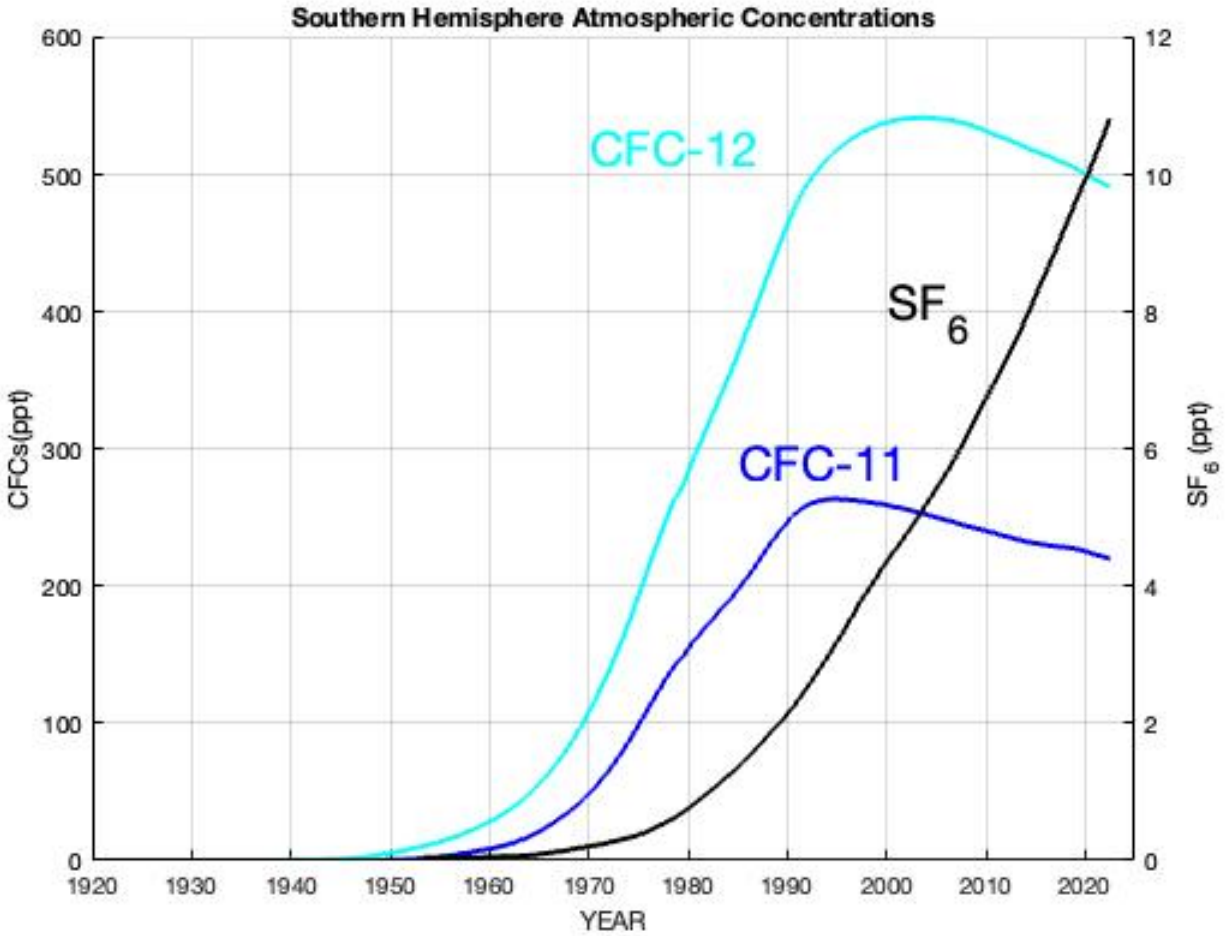

Fig. 8. Mean annual tropospheric CFC-11, CFC-12 and sulfur hexafluoride (SF_6) concentrations in the southern (SH) hemisphere for the period 1920.5 to 2022.5. The concentrations are expressed as the mixing ratio (mole fraction) of the trace gas in dry air and are reported in parts-per-trillion (ppt).

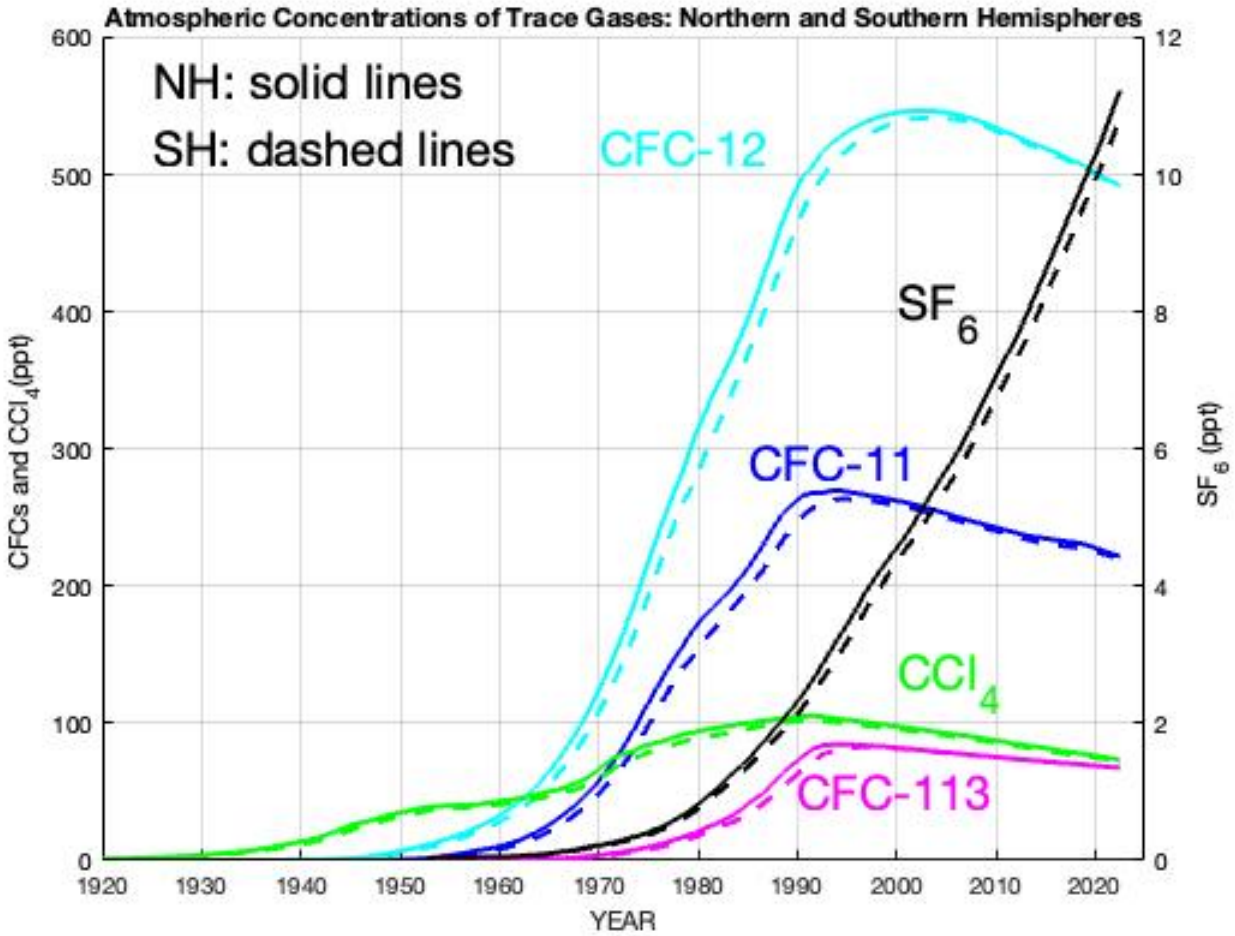

Fig. 9. Mean annual tropospheric CFC-11, CFC-12, CFC-113, carbon tetrachloride (CCl_4), sulfur hexafluoride (SF_6) concentrations in the northern (NH) and southern (SH) hemispheres for the period 1920.5 to 2022.5. The concentrations are expressed as the mixing ratio (mole fraction) of the trace gas in dry air and are reported in parts-per-trillion (ppt).